# Om Jaiswal

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## Education

**Vellore Institute of Technology** — Bhopal, MP, India
*Bachelor of Technology in Computer Science Engineering; GPA: 8.63* — *August 2023 - July 2027*

***Relevant Coursework:*** *Object Oriented Programming, Databases, Linux, Data Structures and Algorithms, Operating Systems, Computer Architecture.*

## Skills Summary

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|---|---|
| **Specialisation:** | Data Structure and Algorithms, Data analytics |
| **Languages:** | Python, C++, Java, C, JS |
| **Database:** | PostgreSQL, MySQL, SQLite |
| **Platforms:** | Ubuntu, Windows |

## Projects

**Plant Disease Detection** | *Python, TensorFlow, OpenCV, Keras, face recognition*

- Developed a deep learning-based plant disease detection system achieving **90%+ accuracy** using **TensorFlow** and **Keras**, significantly reducing manual inspection time for agricultural applications
- Architected a custom **CNN architecture** with image preprocessing via **OpenCV**, enhancing model performance on diverse leaf images under varying lighting conditions
- Optimized model performance through advanced techniques including data augmentation, transfer learning, and hyperparameter tuning
- Trained on a dataset of **10,000+ annotated leaf images** covering multiple plant species and disease types

**Sleep Apnea Detection System** | *Python, Librosa, noisereduce, Matplotlib*

- Crafted an audio-based sleep disorder detection system analyzing respiratory patterns, achieving **85%+ accuracy** in classifying apnea types from breath sounds using **librosa** for feature extraction
- Applied advanced signal processing techniques including **spectral noise reduction** (noisereduce), **zero-crossing rate analysis**, and **time-frequency transformations** for robust feature extraction
- Designed an interactive visualization system with **Matplotlib** displaying waveforms, spectrograms, and acoustic features in real-time for medical analysis
- Developed a category-based exploration system classifying **Hypopnea**, **Obstructive Apnea**, and **Mixed Apnea** events with timestamped audio segments

**Sentiment Analysis System** | *PyTorch, NLP, TF-IDF, NLTK*

- Created an end-to-end sentiment classification system using PyTorch that achieved 85%+ accuracy in binary sentiment prediction (positive/negative) on textual data
- Formulated an advanced NLP preprocessing pipeline with **NLTK**, including tokenization, stopword removal, and text normalization, reducing noise in input data by 40%
- Devised feature extraction using **TF-IDF vectorization** (scikit-learn) to transform raw text into optimized numerical representations for neural network input
- Built a production-ready prediction pipeline capable of processing user input in real-time with **500ms latency** from raw text to sentiment classification
- Executed batch training (batch size=16) with PyTorch **DataLoader** for efficient memory management during model training on CPU/GPU

**Face Recognition System** | *Python, OpenCV, dlib, Flask, SQLAlchemy*

- Spearheaded the creation of a real-time face recognition system using **dlib** and **face-recognition** libraries, achieving **95%+ accuracy** in identity verification under varying lighting conditions
- Incorporated advanced **liveness detection** through blink pattern analysis, reducing spoofing attempts by 80% compared to basic face matching
- Built a secure **Flask** web interface with user management capabilities including registration, authentication, and profile management
- Integrated **SQLAlchemy** for database operations, ensuring secure storage of facial encodings and user credentials with proper encryption

## Co-Curricular Activities

**IBM AI Foundations** | *IBM SkillsBuild*

- Earned IBM's "Getting Started with Artificial Intelligence" certification, mastering fundamental AI concepts including machine learning workflows, neural networks, and ethical AI principles.

**Immersion, P&T, VIT Bhopal** | *Finalist*

- Finalist in a competitive coding immersion event organized by the Placement & Training Department, securing a position among the top performers in the college.

**Data Science Certification** | *MASSAI, IIT Guwahati* | *Expected Dec 2025*

- Mastering machine learning algorithms including regression, classification, and neural networks through industry-grade projects
- Building data analysis pipelines using Python (Pandas, NumPy) and visualization tools (Matplotlib, Tableau)
- Deploying big data solutions with SQL and NoSQL databases for real-world business case studies
- Constructing end-to-end data science applications with model deployment using Flask and cloud platforms